

PURPOSE AND CURRENT POSITION

Own the engine. Prove the first use case.

An original C/C++ deep-learning stack, first demonstrated through CT segmentation of kidney, tumor and cyst.

What exists today

- CT viewer size restriction fixed for the 522 MiB scan.
- Native training model expanded to 1.55 million parameters.
- Tesla T4 execution verified: 52/52 regressions plus a small-model CUDA correctness and resume gate.
- 64 native CPU updates fitted one real patch; the live two-case research demo is verified.

What is still unproven

- Full-size native updates still diverge from the PyTorch reference.
- Current patch results do not establish whole-scan accuracy or useful segmentation quality.
- GPU path remains hybrid; speed and larger runs need measured validation.
- Native adversarial GAN and production application are incomplete.

TODAY: A TRANSPARENT RESEARCH DEMONSTRATION

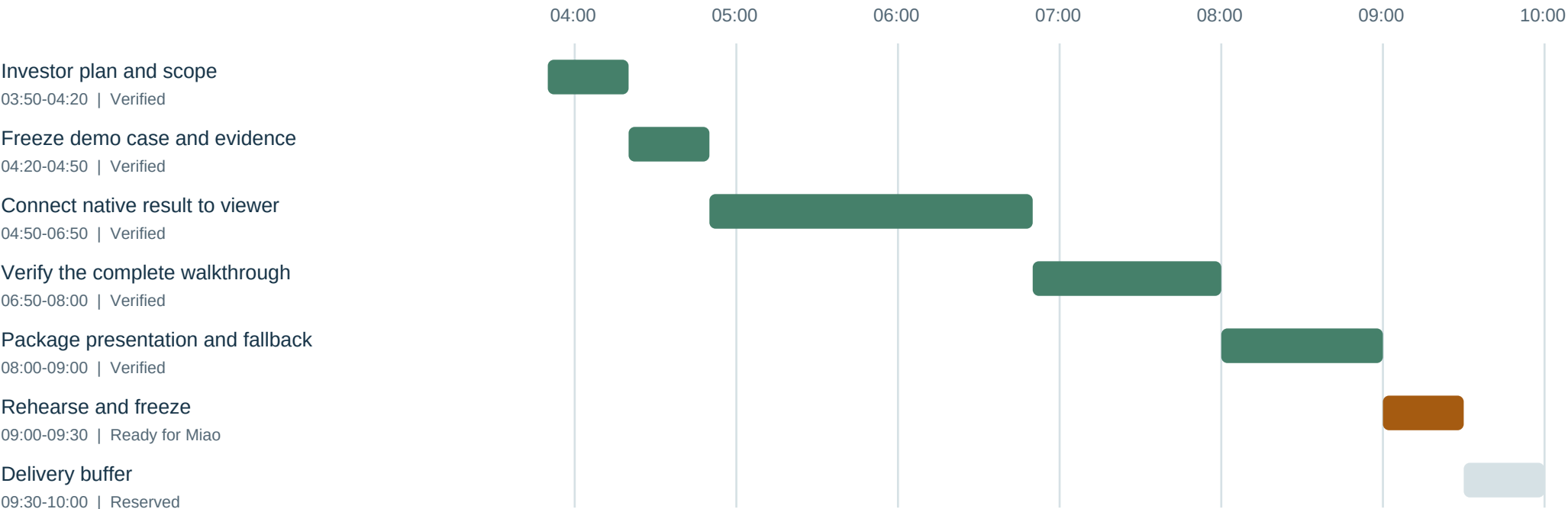
Show the selected CT, reference mask, genuine native output and recorded training evidence. Keep model limitations visible. No production-readiness or clinical-performance claim.

Current focus: reference drift and held-out quality. The end-user website update is live.

TODAY'S DELIVERY

10:00 a.m. today is the hard deadline.

Wednesday 30 September 2026. Planned work windows below; first five deliverables verified ahead of schedule. Miao owns the rehearsal and presentation.



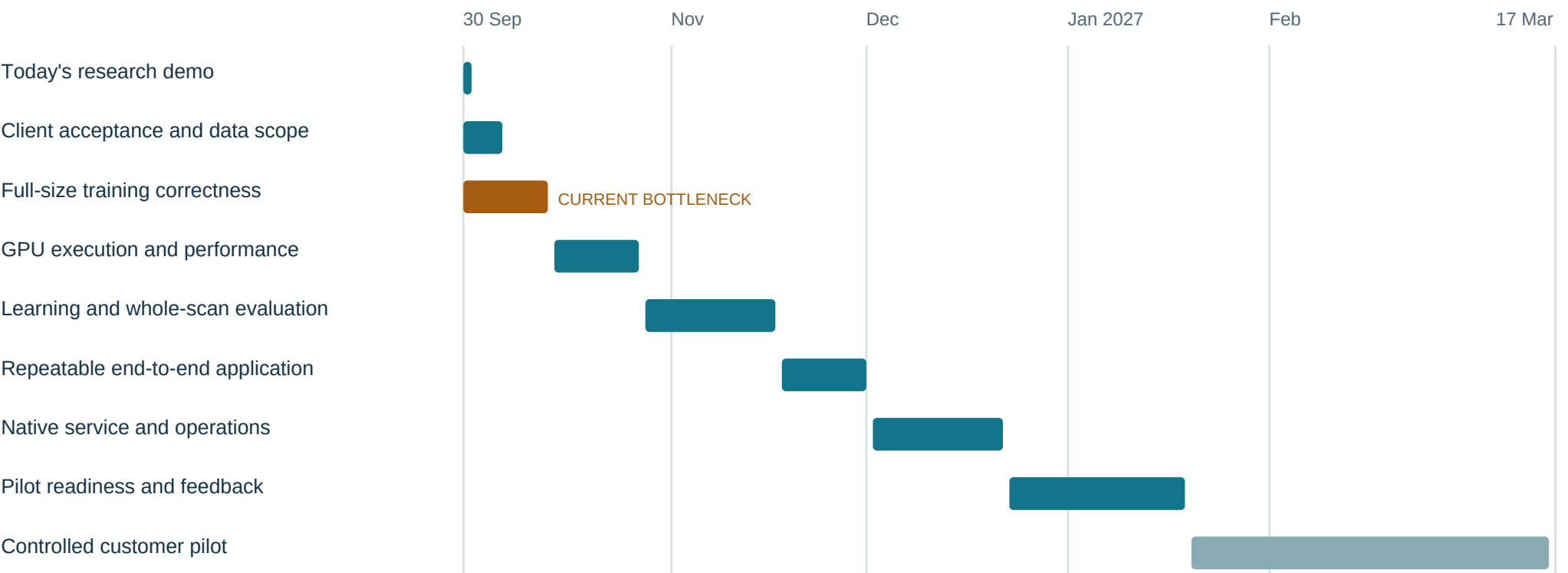
Definition of done: input, reference and native result are distinguishable; crop versus whole scan and stored versus live execution are labelled; the walkthrough launches cleanly; logs and limitations are available.

Current position: the bounded live demo, offline fallback and public website are verified. Ready for Miao's rehearsal. The 9:30 freeze and 10:00 delivery buffer remain reserved; whole-scan quality is a later gate.

WHOLE-PROJECT GANTT

From proof of execution to a customer pilot.

Baseline estimates after today. A failed acceptance gate moves every dependent date. No paid compute, extra team or continuous background work is assumed.



2 Dec: application review | 20 Jan: pilot go/no-go | 17 Mar: pilot outcomes review

Critical path: numerical correctness -> full-size GPU validation -> whole-scan learning -> repeatable application -> pilot readiness -> pilot. Client acceptance work proceeds alongside diagnosis. Today's bounded demonstration is separate from the later quality-qualified application.

EXECUTION AND INVESTMENT DISCIPLINE

Fund the next proof, then expand the platform.

Client needs first. Original C/C++ implementation second. Remaining historical backup recovery last; retain evidence from every new run.

Gates that control delivery

- **Correctness:** same inputs and weights; gradients, moments and updates pass existing tolerances; resume verified.
- **Quality:** frozen patient splits; separate per-class and grouped metrics; whole scans and failure cases. Acceptance thresholds agreed with Miao by 6 October.
- **Application:** correct overlays, results saved, jobs recover, errors are useful; agreed latency measured on named hardware.
- **Pilot:** named partner, authorised data, independent review and support process. No clinical deployment date is promised.

Resources and longer horizon

- **Miao:** client scope, investors, data access, partner and spending decisions. **Codex:** implementation and evidence during active work.
- GPU hours and costs follow measured throughput. Free-session availability is a schedule risk; extra staffing is not assumed.
- **6-12+ months, indicative:** general graph/autodiff, broader operators, native orchestration, additional formats, multi-GPU and representation learning.
- GAN/adversarial training follows a verified supervised baseline. Broader foundation-model capability needs a separate dataset, budget and evaluation plan.

Reporting: proposed weekly investor update with deliverable, evidence, gate, next action and changed dates. Rebaseline after the first numerical diagnosis. This schedule is not an automation or a promise of unattended execution.

Evidence basis: repository CI and saved Kaggle Version 1, verified as of 30 September 2026. Full detail is retained in the project evidence; private source, credentials and checkpoint internals are excluded here.